



Analysis of the Laundry Web Application Using the CIA Triad Approach

Group 11

Names : 1. Mufti Shafiq (2421010104)
2. Nazilatun Natasya Ramadhani (2421010173)

Class : 3CS1

Faculty : 1. Ivan Firdaus
2. Tri Agus Riyadi, S.Kom., M.Kom.

CEP CCIT
FAKULTAS TEKNIK UNIVERSITAS INDONESIA
2025

PROJECT INFORMATION

Project Title : Analysis of the Laundry Web Application Using the
CIA Triad Approach

Batch Code : 3CS1

Start Date : October 1, 2025

End Date : October 17, 2025

Names of Faculty : Mr Ivan Firdaus

Mr Tri Agus Riyadi, S.Kom., M.Kom.

Names of Developer : Mufti Shafiq

Nazilatun Natasya Ramadhani

Date of Submission: October 17, 2025

ACKNOWLEDGEMENT

Praise our gratitude to Allah SWT, because for the abundance of grace the authors can complete this paper on time without any significant obstacles and in accordance with expectations.

Our gratitude goes to Mr. Ivan Firdaus, and Mr. Tri Agus Riyadi, S.Kom., M.Kom. as a class lecturer and supervisor who has helped provide direction and understanding in the preparation of this paper.

We realize that in the preparation of this paper there are still many shortcomings due to our limitations. Therefore, the compiler really hopes for criticism and suggestions to improve this paper. Hopefully what is written can be useful for all parties in need.

SYSTEM SUMMARY

This laundry management system is a simple web-based application designed to help users manage customer data and service transactions more efficiently. The system allows users to log in, add customer information, and view basic records related to laundry services. It was developed using Python (Flask framework) and MySQL for database management, with HTML templates for the interface.

The main goal of this system is to demonstrate how web applications interact with databases to perform basic CRUD (Create, Read, Update, Delete) operations. Although the project focuses on functionality and structure rather than security, it provides a good foundation for understanding how authentication and data handling work in a small-scale environment.

Through this project, students can observe how a simple HTTP-based system operates, and analyze its strengths and weaknesses using the CIA (Confidentiality, Integrity, Availability) framework to identify potential improvements in data protection and system reliability.

SYSTEM ANALYSIS

Case Study: Laundryy Web Application

This case study discusses the Laundryy web application, a web-based system developed using the Flask framework and MySQL database. The application is designed to manage customer information, transactions, and authentication processes efficiently. The main objective of this study is to analyze the security level of the system using the CIA Triad framework, which consists of Confidentiality, Integrity, and Availability principles.

CIA Triad Analysis

The system was evaluated based on the three aspects of the CIA Triad to identify vulnerabilities in data management and operational reliability.

The findings are summarized as follows:

Aspect	Findings In Project	Explanation
Confidentiality	Password stored in plain text and database uses root user without password.	User data is not fully protected. Anyone who gets access to the database can see all usernames and passwords easily.
Integrity	No validation when adding or updating customer data.	Data can be changed or deleted without control, which may cause wrong information in the system.
Availability	App runs in debug mode and has no backup system.	If there is an error or crash, the system can stop working and data might be lost.

DISPLAY

To improve the Confidentiality aspect identified in the analysis, password hashing was implemented in the login module. Instead of storing passwords in plain text, the system now uses a hashing algorithm to protect credentials in the database.

The updated user interface is shown below.

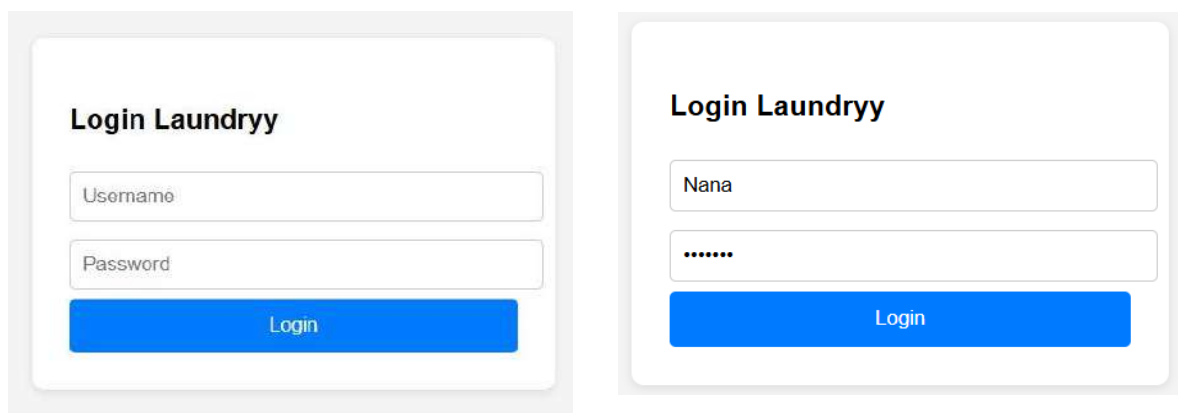
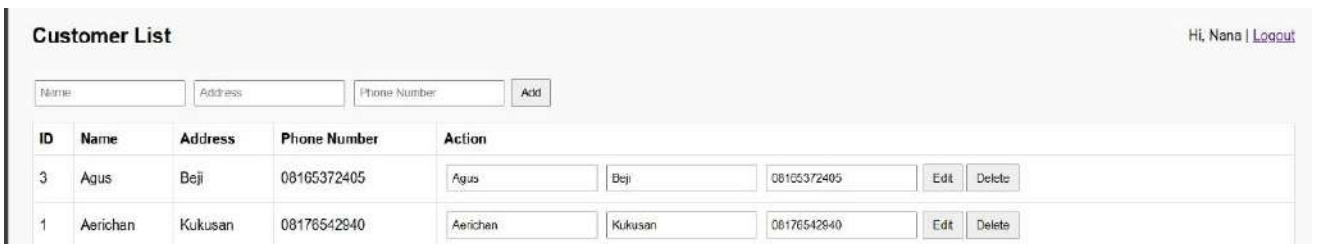


Figure 1. Login Display After Password Hashing Implementation

After logging in, users can view and manage customer data.



ID	Name	Address	Phone Number	Action
3	Agus	Beji	08165372405	Agus Beji 08165372405 Edit Delete
1	Aerichan	Kukusan	08176542940	Aerichan Kukusan 08176542940 Edit Delete

Figure 2. Customer Data Display

SCRIPT

Wireshark Network Traffic Analysis

To further analyze the security of data transmission between the client and the server, a network traffic analysis was performed using Wireshark. Wireshark is a packet-sniffing tool used to capture and inspect data packets that travel through a network interface in real time. This test aimed to determine whether user credentials, particularly usernames and passwords, were properly protected during the login process.

The first step was selecting the active network interface, which in this test was the Wi-Fi connection used by the system.

The steps of the analysis are as follows:

1. Selecting the Network Interface

The first step was to open Wireshark and select the Wi-Fi interface used by the device running the Laundryy web application. This interface was chosen because the client accessed the application through a local wireless network.

By capturing packets on this interface, it was possible to monitor all HTTP requests and responses between the client browser and the web server.

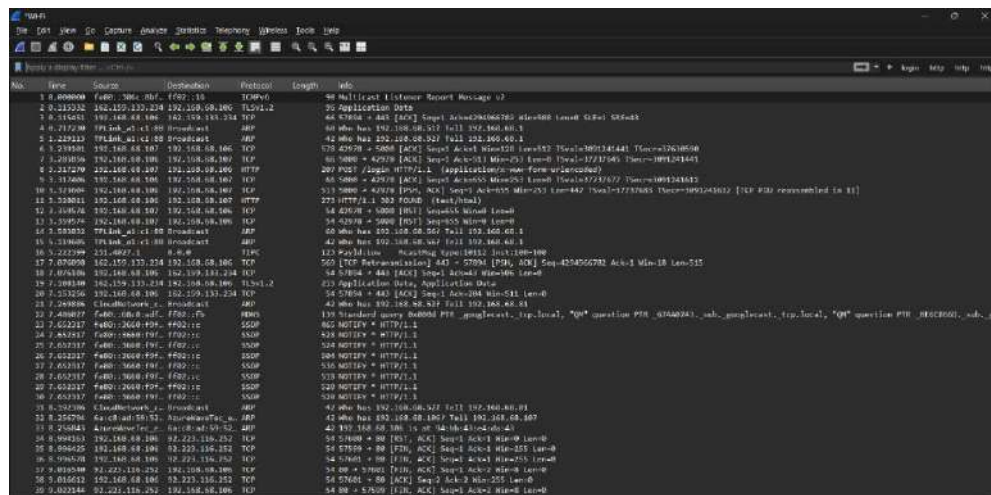
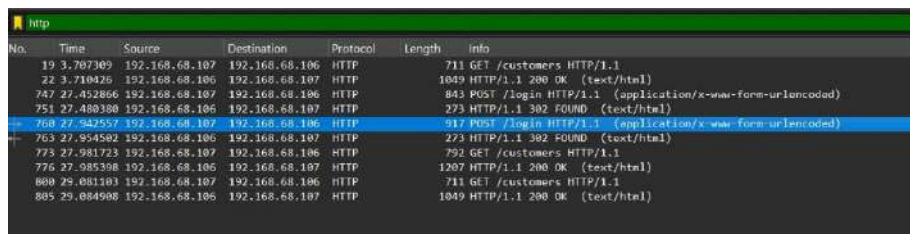


Figure 3. Selecting Wi-Fi Interface in Wireshark

SCRIPT

2. Applying the HTTP Filter

After starting the capture, a display filter http was applied in Wireshark. This filter allowed the tool to show only HTTP packets, excluding other network protocols that were not relevant to this analysis. Since HTTP transmits data in plaintext without encryption, it is suitable for checking whether sensitive information such as login credentials can be exposed.



The image shows the Wireshark packet list with the filter 'http' applied. The list displays only HTTP packets. The selected packet is number 760, which is an HTTP POST request to /login.

No.	Time	Source	Destination	Protocol	Length	Info
19	3.707309	192.168.68.107	192.168.68.106	HTTP	711	GET /customers HTTP/1.1
22	3.710426	192.168.68.106	192.168.68.107	HTTP	1049	HTTP/1.1 200 OK (text/html)
747	27.452866	192.168.68.107	192.168.68.106	HTTP	843	POST /login HTTP/1.1 (application/x-www-form-urlencoded)
751	27.480380	192.168.68.106	192.168.68.107	HTTP	273	HTTP/1.1 302 FOUND (text/html)
760	27.942557	192.168.68.107	192.168.68.106	HTTP	917	POST /login HTTP/1.1 (application/x-www-form-urlencoded)
763	27.954502	192.168.68.106	192.168.68.107	HTTP	273	HTTP/1.1 302 FOUND (text/html)
773	27.981723	192.168.68.107	192.168.68.106	HTTP	792	GET /customers HTTP/1.1
776	27.985398	192.168.68.106	192.168.68.107	HTTP	1207	HTTP/1.1 200 OK (text/html)
800	29.001103	192.168.68.107	192.168.68.106	HTTP	711	GET /customers HTTP/1.1
805	29.004908	192.168.68.106	192.168.68.107	HTTP	1049	HTTP/1.1 200 OK (text/html)

Figure 4. Applying HTTP Filter in Wireshark

3. Observing the Login Request (HTTP POST)

During the capture, a login attempt was performed on the Laundryy application. In the captured traffic, an HTTP POST request was found containing the user's input data. Upon inspection of the packet contents, the username and password fields were clearly visible in plaintext. This shows that the credentials were transmitted without encryption from the client to the server.

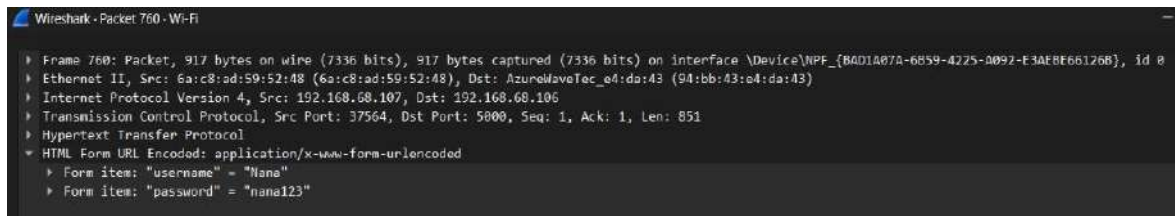


Figure 5. HTTP POST Request Showing Username and Password in Plaintext

From the captured packets, it was found that the username and password were still visible in the HTTP POST request. This means the system transmits login data without encryption, which could allow attackers to intercept sensitive information.

SYSTEM CONFIGURATION

Hardware :

1. HP

Operating System:

1. Windows Server 64-bit

Software :

1. Pycharm
2. MySQL
3. Ms. Word
4. Power Point
5. Github : (<https://github.com/nacnachaa/project-2>)

PROJEK FILE DETAILS

No	FileName	Remarks
1.	3CS1 PROJECT 2 .pdf	Paper File
2.	Project 2 Presentation .pptx	Presentation File